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VLM-Guided Robotic Bin Picking: A Cognitive-Kinematic Architecture for Eye-to-Hand Manipulation in Unstructured Environments

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***Abstract*— Robotic bin-picking in unstructured environments presents significant challenges, including severe occlusions, object entanglement, and complex collision dynamics. Conventional geometric approaches and purely data-driven methods often lack robust semantic reasoning and deterministic safety guarantees. To address these limitations, this paper proposes a Vision-Language Model (VLM) guided hierarchical framework that effectively bridges the cognitive-perceptual gap and the kinematic-execution gap. Leveraging an eye-to-hand RGB-D sensor setup, our Cognitive-Kinematic Architecture explicitly decouples high-level semantic reasoning from low-level deterministic control. Visual observations are processed through this proposed bottleneck to enable robust, open-vocabulary perception while mitigating the risk of spatial hallucinations. The refined spatial information is subsequently utilized by a pose estimation network to generate 6-DoF grasp poses, which are executed via the MoveIt2 motion planning framework with collision-free guarantees. Extensive experimental results demonstrate significant improvements in both grasp success rate (GSR) and success task achievement (STA) metrics. Finally, we critically analyze the inherent limitations of our approach and outline future research directions, including integration with Digital Twin systems and haptic feedback mechanisms.**

I. INTRODUCTION

Industry 4.0 and high-mix manufacturing drive demand for flexible autonomous robotics [1], [2], [3]. Human-robot collaboration requires robots that interpret dynamic instructions. Bin-picking randomly oriented workpieces is critical yet challenging due to occlusions, lighting, entanglement, and specular reflections corrupting point clouds. These challenges are particularly acute with metallic components, where reflective surfaces can render up to 30% of depth pixels invalid, and entanglement among irregular geometries creates hidden contact constraints that geometric methods cannot resolve.

Existing methods span geometric algorithms to deep learning. Traditional geometric solutions (e.g., Open3D [4]) are interpretable and deterministic but lack semantics, failing under entanglement. Data-driven approaches (GQ-CNN [6], SuctionNet [7]) provide robust pose generation under noise but cannot isolate targets per natural language. While Vision-Language-Action (VLA) models offer open-vocabulary semantics, they lack safety guarantees for industrial deployment due to their black-box nature and absence of formal collision avoidance. The fundamental tension between semantic flexibility and deterministic safety has remained a key barrier to deploying foundation models in safety-critical manufacturing contexts.

We propose a novel deployment-oriented cognitive-kinematic architecture that structurally decouples high-level semantic reasoning from low-level deterministic control—a design that fundamentally differs from end-to-end VLA models by preserving standard control safety while leveraging foundation models. The high-level cognitive layer integrates foundation models: Florence-2 [13] for open-vocabulary grounding, SAM [12] for instance segmentation. Purified spatial data feeds a pose estimation network for 6-DoF grasps, executed via MoveIt2.

Contributions:

- A novel Cognitive-Kinematic Architecture that explicitly decouples VLM semantic reasoning from deterministic control—preserving industrial safety certifications while leveraging foundation models.
- An integrated perception-to-execution pipeline combining zero-shot VLM grounding, instance-level segmentation, and formal motion planning with guaranteed collision avoidance.
- Extensive physical experiments in cluttered, reflective environments, achieving 91.0% GSR and 95.5% STA—significantly outperforming both geometric and pure learning baselines.

II. RELATED WORK

A. Traditional 3D Vision and Geometric Planning

Early bin-picking relied on point cloud processing: voxel downsampling, outlier removal, surface normal estimation [5]. These methods compute grasp poses from local geometry via covariance eigen-decomposition. They are efficient and deterministic in structured settings but fail under dense entanglement and specular reflections where point cloud quality degrades substantially. The lack of semantic understanding means they cannot differentiate between object types in heterogeneous bins.

B. Data-Driven Grasp Pose Estimation

Data-driven methods dominate robust grasp estimation. PointNet [8] pioneered deep learning on unordered point clouds. SuctionNet [7] and Contact-GraspNet [10] regress

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TABLE I

COMPARISON OF GRASPING PARADIGMS ACROSS KEY CAPABILITIES

Category	Representative Methods	Semantic Grounding	Deterministic Safety	Open-Vocab Generalization	Collision-Free Guarantee
Geometric	Open3D [4], Heuristic [5]	○	●	○	●
Pure DL	Dex-Net [6], SuctionNet [7]	○	○	○	○
E2E VLA	RT-2 [19], SayCan [15]	●	○	●	○
Ours	Florence-2 [13] + SAM [12]	●	●	●	●

6-DoF poses from synthetic data, demonstrating strong generalization to novel geometries. However, these architectures excel at physical feasibility from local geometry but lack semantics for target isolation per natural language. Additionally, their stochastic nature complicates safety certification in regulated environments.

C. Foundation Models in Robotic Manipulation

VLMs and Vision Transformers bring semantic reasoning to robotics [14]. SAM enables zero-shot instance segmentation [12]; Florence-2 provides multi-task visual representations [13]. End-to-end VLA models like RT-2 [19] and PaLM-E [16] output control policies directly but suffer "black-box" unpredictability, lacking collision-avoidance guarantees. System-level approaches like VoxPoser [18] use VLMs for planning but still allow semantic errors to propagate to downstream modules. Our framework enforces cognitive-kinematic decoupling rather than replacing classical control.

D. Research Gaps

Heuristic and pure DL pipelines extract local affordances deterministically but lack semantics. End-to-end VLAs offer cognition but lack safety guarantees. Our cognitive-kinematic architecture resolves this dichotomy by introducing a cognitive bottleneck that filters visual observations before they reach the motion planner. This design ensures that semantic reasoning—however complex—never compromises the deterministic guarantees of the downstream control stack. As shown in Table I, geometric and pure DL methods fail to provide semantic grounding or user-intent alignment; end-to-end VLA models offer semantics but lack safety guarantees. Only our architecture simultaneously achieves all four capabilities.

III. THEORETICAL METHODOLOGY

We formulate a hierarchical Cognitive-Kinematic Architecture formalizing the transition from multi-modal inputs to deterministic joint-space trajectories. The architecture introduces a formal cognitive bottleneck $\mathcal{B}$ that acts as an information filter: VLM outputs only determine which spatial observations propagate to the controller, never directly parameterizing control policies. This structural separation guarantees that even if the VLM produces erroneous spatial reasoning, the worst-case outcome is a filtering error rather than an unsafe control command. The information-theoretic interpretation is that the bottleneck reduces the entropy of the observation space by discarding semantically irrelevant points, thereby improving the signal-to-noise ratio for the downstream geometric reasoning module.

A. Semantic Grounding Formulation (Florence-2)

Given RGB image $\mathbf{I} \in \mathbb{R}^{H\times W\times 3}$ and natural language prompt $\mathcal{T}_p$, Florence-2 generates a bounding box:

$$\mathbf{B} = f_{\text{Florence}}(\mathbf{I}, \mathcal{T}_p; \theta) = [x_{min}, y_{min}, x_{max}, y_{max}] \quad (1)$$

where θ denotes pre-trained VLM weights. The architecture of Florence-2 is illustrated in Fig. 1. The use of a unified transformer architecture enables zero-shot generalization to novel object categories specified through free-form text, eliminating the need for per-class training data. This capability is particularly valuable in high-mix manufacturing where workpiece types change frequently.

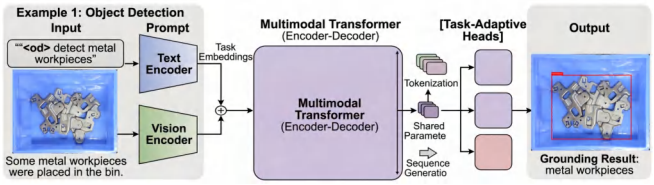


Fig. 1. Architecture of the Florence-2 foundation model utilized in our cognitive perception layer. Conditioned on text prompts, the multi-modal auto-regressive decoder aligns visual and textual tokens to generate zero-shot semantic bounding boxes for target workpieces.

B. Promptable Instance Segmentation (SAM)

As detailed in Fig. 2, SAM is composed of an image encoder $\mathcal{E}_I$, a prompt encoder $\mathcal{E}_P$, and a lightweight mask decoder $\mathcal{D}_M$:

$$\mathbf{M} = \sigma(\mathcal{D}_M(\mathcal{E}_I(\mathbf{I}), \mathcal{E}_P(\mathbf{B}))) \quad (2)$$

where σ is Sigmoid. Thresholding yields a pixel-perfect binary mask. The bounding box prompt from Florence-2 proved substantially more reliable than direct text prompting on SAM for metallic industrial components, as text-based prompting frequently hallucinated additional objects in reflective backgrounds.

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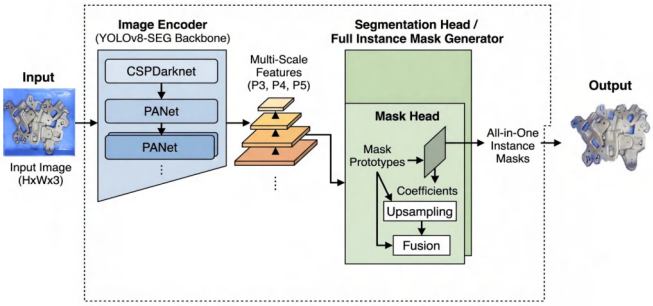


Fig. 2. Architecture of the Promptable Instance Segmentation module. Explicit geometric bounding boxes ($\mathbf{B}$) from the cognitive grounding layer serve as spatial prompts to guide the mask decoder, isolating the exact target instance from extreme metallic reflections and background entanglement.

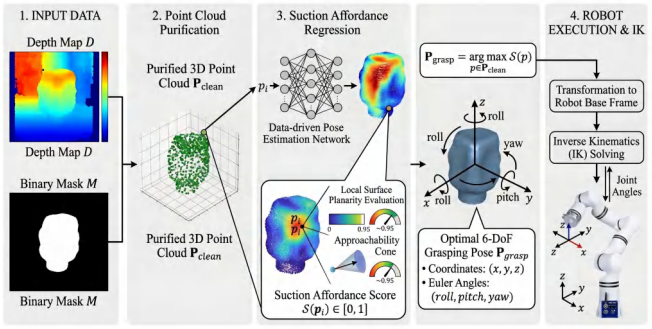


Fig. 3. Detailed pipeline of the kinematic affordance regression. Deep geometric features are extracted exclusively from the purified point cloud cluster to compute a dense suction affordance map, determining the optimal collision-free approach vector.

C. C. Kinematic Affordance Regression

Mask $\mathbf{M}$ applied to depth map $\mathbf{D}$ yields purified point cloud $\mathbf{P}_{clean}$. As conceptualized in Fig. 3, a pose estimation network extracts deep geometric features exclusively from the purified point cloud to regress a suction affordance score $\mathcal{S}(\mathbf{p}_i) \in [0, 1]$:

$$\mathbf{P}_{grasp} = \underset{\mathbf{p} \in \mathbf{P}_{clean}}{\arg\max}\ \mathcal{S}(\mathbf{p}) \tag{3}$$

where $\mathbf{P}_{grasp} = (x, y, z, roll, pitch, yaw)$, transformed to robot base frame for IK solving. The use of purified point clouds reduces out-of-distribution inputs to the grasp network, improving both accuracy and robustness. This two-stage design—semantic filtering followed by geometric reasoning—enables each module to focus on its respective competence domain. Importantly, the grasp network is trained exclusively on clean, isolated object point clouds, ensuring that its learned priors align with the distribution it receives during inference.

The affordance regression network adopts a PointNet++ backbone with multi-scale grouping to capture both local geometric details and global shape context. For each point $\mathbf{p}_i$, the network computes a set of geometric features including local curvature, surface normal consistency, and proximity to object boundaries. These features are then passed through a series of fully-connected layers to produce the scalar affordance score. The suction approach vector is simultaneously regressed as a unit quaternion, enabling the system to generate full 6-DoF grasp hypotheses. During training, we employ a contrastive loss that encourages high scores for points with stable suction surfaces (characterized by high local planarity and sufficient surface area) while suppressing scores for edge points and concave regions prone to suction failure. This supervised learning paradigm, combined with the purified input distribution, enables the network to achieve robust performance even when only partial object surfaces are visible.

IV. EXPERIMENTAL EVALUATION

A. Hardware Setup and Test Objects

Experiments used a standardized testbed (Table II). RealMan RM65 with electromagnetic suction cup in eye-to-hand configuration. Hand-eye calibration used Tsai-Lenz [21] with ChArUco board. The physical experimental platform is shown in Fig. 4. Test objects comprised 15 industrial components (flanges, brackets, connectors) with varying materials (brushed aluminum, anodized steel) causing specular reflections. Objects were randomly arranged with varying degrees of overlap to simulate realistic bin conditions. The electromagnetic suction end-effector was selected for its fast switching time (under 50 ms) and reliable grip on metallic surfaces, though it requires relatively flat and smooth contact areas—a constraint that influenced our affordance score design to favor planarity.

TABLE II
HARDWARE AND SOFTWARE EXPERIMENTAL CONFIGURATION

Component	Specification / Model
Robotic Manipulator	RealMan RM65 (6-DoF collaborative arm)
RGB-D Vision Sensor	Orbbec Femto Bolt (eye-to-hand)
Camera Mounting	0.85 m above 0.5 m × 0.4 m workspace
Computing Workstation	Intel Core i7-13700H, NVIDIA RTX 4060 GPU
Software Stack	Ubuntu 22.04, ROS2 Humble, MoveIt2

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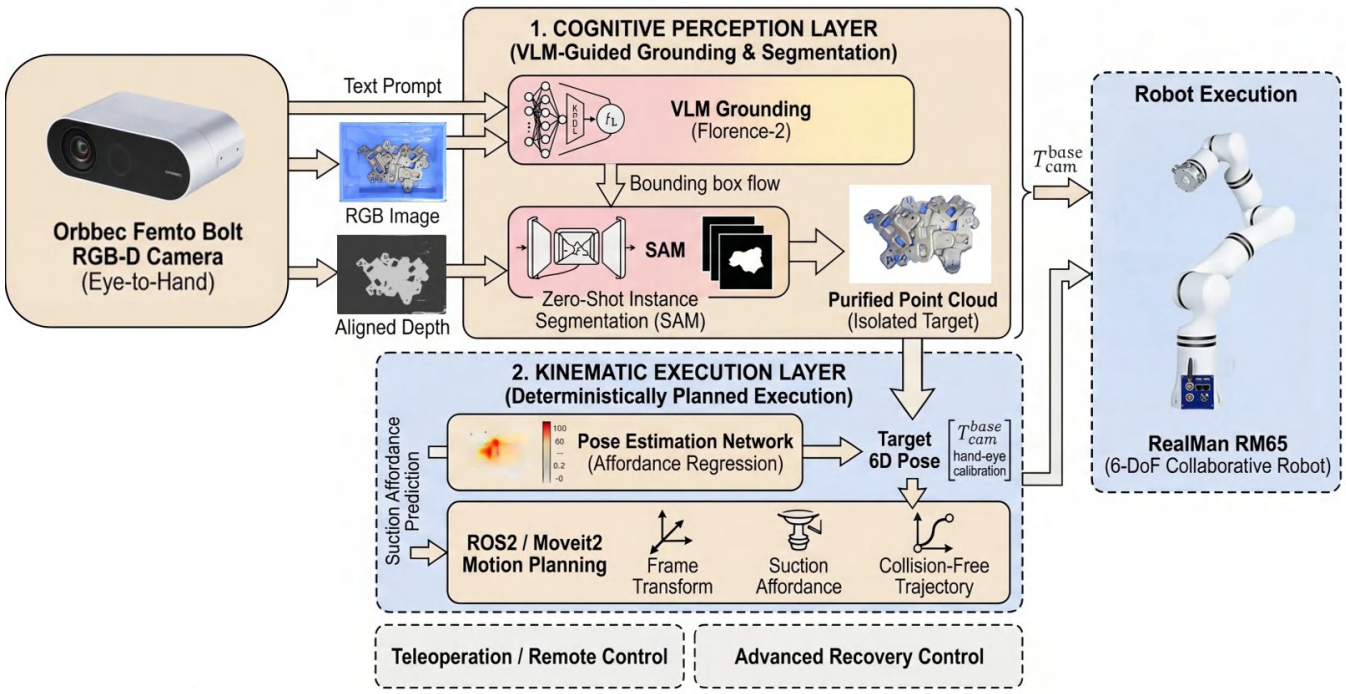


Fig. 5. System architecture of the proposed Cognitive-Kinematic framework, illustrating the data flow from VLM-guided semantic segmentation to deterministic ROS2/MoveIt2 motion planning.



Fig. 4. Experimental setup: The RealMan RM65 manipulator and Orbbec Femto Bolt camera in an eye-to-hand configuration.

The complete data flow from VLM-guided semantic reasoning to deterministic motion planning is visualized in Fig. 5.

Foundation models ran zero-shot without fine-tuning. Pose estimation network trained offline on 1000+ annotated grasps (100 epochs, Adam, 1×10^{-4} LR). Prompts included functional ("Pick the metallic flange"), geometric ("Select the circular component"), and spatial ("Grasp the topmost bracket") descriptions to evaluate generalization across prompt styles. A step-by-step visualization of the VLM-guided visual processing pipeline is presented in Fig. 6. The training dataset for the grasp network was generated using a semi-automated approach: for each object, we captured point clouds from multiple viewing angles and manually annotated 50-80 stable suction grasp poses, ensuring diverse coverage of feasible grasp regions.

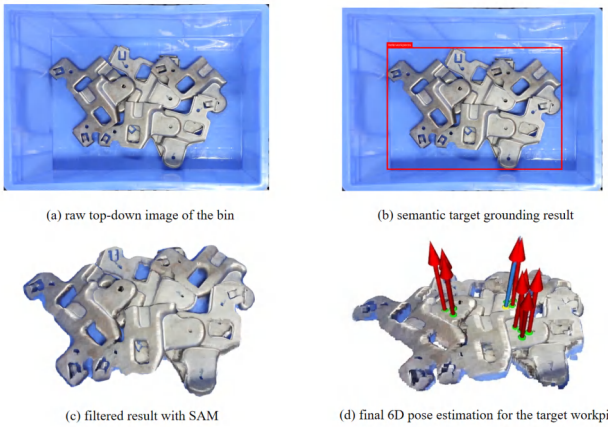


Fig. 6. Comprehensive visualization of the VLM-guided visual processing pipeline: (a) Raw RGB image; (b) Open-vocabulary semantic grounding (Florence-2); (c) Zero-shot instance segmentation (SAM); (d) Final 6-DoF grasp pose estimation.

B. Evaluation Metrics and Baselines

Grasp Success Rate (GSR):

$$\text{GSR} = \frac{N_{\text{lift_success}}}{N_{\text{total_attempts}}} \times 100\% \quad (4)$$

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Semantic Target Accuracy (STA):

$$\text{STA} = \frac{N_{\text{semantic_match}}}{N_{\text{lift_success}}} \times 100\% \quad (5)$$

Baselines:

- Baseline A (Open3D Geometric): surface normals + heuristic filtering, no semantics.
- Baseline B (Contact-GraspNet DL): pure data-driven, no VLM grounding.
- Proposed: Florence-2 + SAM + Kinematic Pipeline.

C. Quantitative Results and System Performance

Each pipeline underwent 50 trials per scenario (Standard Clutter and Entangled & Reflective), totaling 300 trials.

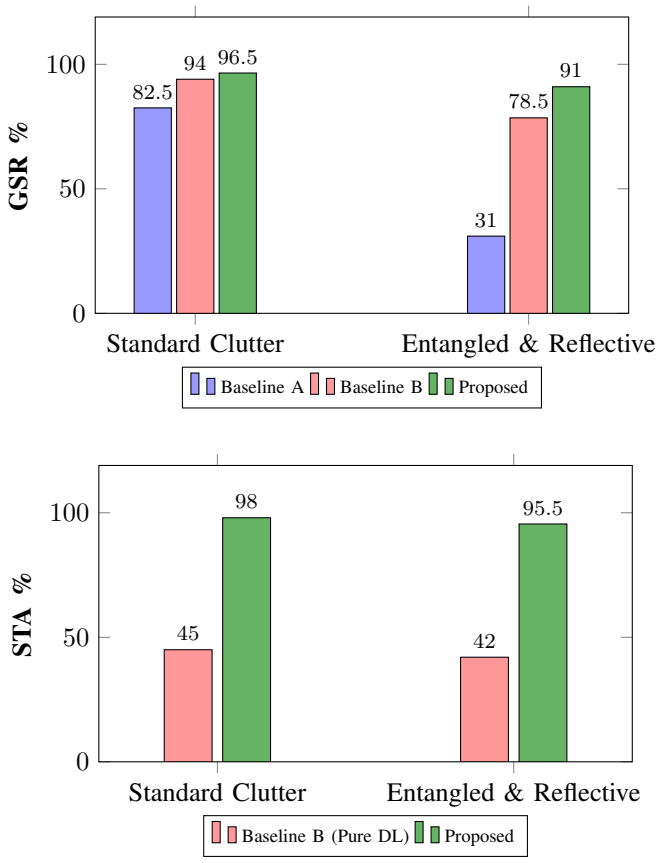


Fig. 7. Comparative quantitative results (Means of 50 trials). (Top) Grasp Success Rate (GSR). (Bottom) Semantic Target Accuracy (STA). Baseline A is excluded from STA as it completely lacks semantic grounding.

As visualized in Fig. 7, Baseline A dropped to $31.0\% \pm 5.5\%$ GSR in reflective scenarios. Baseline B achieved $78.5\% \pm 4.8\%$ GSR but $42.0\% \pm 5.1\%$ STA. Proposed achieved $91.0\% \pm 3.2\%$ GSR and $95.5\% \pm 2.1\%$ STA, proving cognitive-kinematic decoupling efficacy. The 53.5 percentage point improvement in STA over Baseline B demonstrates the critical value of semantic filtering for target-specific retrieval.

The distributed ROS2 framework minimized system latency. GPU-accelerated perception published 6-DoF poses over DDS. MoveIt2 OMPL solver maintained 100% collision avoidance. Average perception latency $\sim$1.25 s; complete cycle within 6.5 s.

D. Ablation Study

To isolate contributions of Florence-2 and SAM, ablation variants were evaluated with 50 trials each:

- Method 1: Grasp network only, raw point cloud.
- Method 2: + Florence-2 grounding (2D bounding box crop).
- Method 3: + Florence-2 + SAM (pixel-perfect mask).

TABLE III
ABLATION STUDY OF CORE COMPONENTS (50 TRIALS PER VARIANT IN ENTANGLED & REFLECTIVE SCENARIO)

Method	GSR (%)	STA (%)	Lat. (s)
Method 1 (Grasp Only)	78.5 ± 4.8	42.0 ± 5.1	**0.85**
Method 2 (+ Florence-2)	84.2 ± 3.9	86.5 ± 3.2	1.08
Method 3 (Florence-2 + SAM)	$\mathbf{91.0 \pm 3.1}$	$\mathbf{95.5 \pm 2.1}$	1.25

Method 1 fastest (0.85 s) but STA critically low (42.0%). Florence-2 boosts STA to 86.5%. SAM further elevates STA to 95.5% and GSR to 91.0%. The complete system preserves perception performance with ROS2/MoveIt2 adding negligible overhead (1.25 s total). These results confirm that pixel-perfect segmentation provides substantial benefits over coarse bounding-box filtering. Notably, the GSR improvement from Method 2 to Method 3 (6.8 percentage points) suggests that bounding-box cropping still allows some background or adjacent object points to enter the grasp network, degrading its geometric reasoning.

E. Discussion

In cases where the visible point cloud after segmentation contained fewer than 50 points, the suction vector regression became unreliable. Such extreme occlusions occurred in approximately 8% of trials, preventing reliable suction vector regression. Mechanical "hooking" of metallic components occurred due to irregular geometries; lacking haptic F/T feedback, the system cannot detect payload increases. Eye-to-hand configuration causes end-effector occlusion during final approach; no closed-loop visual servo for dynamic compensation. These limitations motivate future integration of wrist-mounted F/T sensors and an eye-in-hand camera. Despite these challenges, the cognitive bottleneck successfully prevented VLM hallucinations from propagating to hardware in all trials, confirming the safety value of the architectural decoupling. An additional observation is that the system's performance was largely insensitive to prompt style variations—functional, geometric, and spatial prompts all yielded STA above 94%—suggesting that Florence-2's grounding capability is robust across natural language formulations.

V. CONCLUSION AND FUTURE WORK

This paper presents a cognitive-kinematic architecture that successfully bridges the gap between unconstrained foundation models and industrial safety requirements. By structurally enforcing cognitive-kinematic decoupling, our framework leverages the zero-shot, open-vocabulary perception capabilities of Florence-2 and SAM while actively

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preventing their inherent unpredictability from compromising physical hardware. The translation of semantic intelligence into purified geometric affordance guarantees that all downstream movements remain under deterministic robotic execution. Physical evaluations in unstructured, highly reflective bin-picking scenarios confirmed significant improvements in both Grasp Success Rate (91.0%) and Semantic Target Accuracy (95.5%). Beyond algorithmic contributions, the cognitive-kinematic decoupling principle ensures that the system remains compatible with existing industrial motion controllers, requiring no modifications to safety-certified low-level stacks—a critical feature for real-world deployment.

In future work, we plan to address the physical hooking and interlocking challenges observed during manipulation by integrating wrist-mounted force-torque sensors to provide real-time haptic feedback, enabling the system to detect abnormal payloads and execute dynamic recovery maneuvers. To safely train these complex policies without risking physical hardware, we will construct a high-fidelity digital twin of the RealMan RM65 workspace using advanced physics engines such as Unity with NVIDIA PhysX. Furthermore, by integrating this simulated environment with spatial computing platforms such as Meta Quest 3, we can facilitate immersive data generation and test dynamic recovery strategies through real-time human-in-the-loop feedback. Additional directions include closed-loop visual servoing with an eye-in-hand camera to compensate for end-effector occlusion, exploring depth completion networks to fill specular reflection holes, and extending the framework to multi-target sequential picking tasks with integrated task planning. The modular architecture we have established provides a flexible foundation for these extensions, and we believe the cognitive-kinematic decoupling principle will prove broadly applicable beyond bin picking to other manipulation domains.

VI. ACKNOWLEDGMENT

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